

Visitor Management Backend - Technical Documentation

Spring Boot 4.1.0 · Java 21 · PostgreSQL 17 · Hibernate 7.4

1. Architecture Overview

+-----+ REST Controllers +-----+	
TimeSlotController · VisitorController	
RegistrationController · BriefingController	
EventController · FeedbackController	
EducationalProgramController · BookingController	
+-----+ Services +-----+	
SchedulingServiceImpl · RegistrationServiceImpl	
EventServiceImpl · FeedbackServiceImpl	
EducationalProgramServiceImpl · BookingServiceImpl	
SchedulingRules (constants)	
+-----+ Repositories (JPA) +-----+	
TimeSlotRepository · VisitorRepository	
RegistrationRepository · BriefingRepository	
EventRepository · FeedbackRepository	
SurveySendRepository · TourStopRepository	
WorkshopRepository · AgriActivityRepository	
BookingRepository	
+-----+ JPA Entities +-----+	
TimeSlot · Visitor · Registration · Briefing · Event	
Feedback · SurveySend · TourStop · Workshop	
AgriActivity · Booking	
(extend BaseEntity: id + createdAt + updatedAt)	
+-----+ core/ - shared infrastructure +-----+	
BaseEntity · GlobalExceptionHandler · ApiError	
ResourceNotFoundException · ConflictException	
BusinessException	

Packages:

- core/ - shared infra only (BaseEntity, exceptions, handler)
- modules/visitemanagement/ - Scheduling + Registration sub-modules
- Enums live in the module package (not core/)

2. Database Schema

Entity-Relationship Diagram

+-----+ time_slot +-----+		+-----+ visitor +-----+	
id BIGINT PK		id BIGINT PK	
slot_date DATE NOT NULL		full_name VARCHAR(150)	NOT NULL
start_time TIME NOT NULL		group_size INT	NOT NULL, DEFAULT 1
end_time TIME NOT NULL		email VARCHAR(200)	UNIQUE
max_capacity INT NOT NULL		phone VARCHAR(30)	
status VARCHAR(20) NOT NULL		language VARCHAR(40)	
guide_id BIGINT		visitor_type VARCHAR(20)	NOT NULL
created_at INSTANT NOT NULL		special_needs VARCHAR(500)	
updated_at INSTANT NOT NULL		created_at INSTANT	NOT NULL

```

+-----+-----+ | updated_at INSTANT| NOT NULL
|
| 1 | 1 |
|
| * | 1 | *
+-----+-----+
| registration |
+-----+-----+
| id BIGINT| PK |
| visitor_id BIGINT| FK -> visitor (NOT NULL) |
| time_slot_id BIGINT| FK -> time_slot (nullable, NULL for |
| | event-only registrations) |
| event_id BIGINT| |
| visit_purpose VARCHAR(20)| NOT NULL |
| is_prospect BOOLEAN| NOT NULL, DEFAULT false |
| status VARCHAR(20)| NOT NULL, DEFAULT 'PENDING' |
| created_at INSTANT| NOT NULL |
| updated_at INSTANT| NOT NULL |
| UNIQUE (visitor_id, time_slot_id) |
+-----+-----+
| 1
|
| 1
+-----+-----+
| briefing |
+-----+-----+
| id BIGINT| PK |
| registration_id | FK -> registration |
| | (NOT NULL, UNIQUE) |
| delivered_at INSTANT| |
| staff_member VARCHAR(150) |
| signature VARCHAR(500) |
| status VARCHAR(20)| NOT NULL, 'PENDING' |
| created_at INSTANT| NOT NULL |
| updated_at INSTANT| NOT NULL |
+-----+-----+

```

registration.event_id -> event.id (no FK constraint, soft link)

```

+-----+-----+
| event |
+-----+-----+
| id BIGINT| PK |
| title VARCHAR(150)| NOT NULL |
| event_type VARCHAR(30)| NOT NULL |
| start_date_time TIMESTAMP| NOT NULL|
| end_date_time TIMESTAMP| NOT NULL|
| max_capacity INT| NOT NULL, DEFAULT 10|
| status VARCHAR(20)| NOT NULL, 'DRAFT'|
| description VARCHAR(1000)| |
| location VARCHAR(150)| |
| created_at INSTANT| NOT NULL |
| updated_at INSTANT| NOT NULL |
+-----+-----+

+-----+-----+
| feedback |
+-----+-----+
| id BIGINT| PK |
| visitor_id BIGINT| FK -> visitor (NOT NULL)|
| survey_send_id BIGINT| FK -> survey_send (nullable)|
| origin VARCHAR(20)| NOT NULL |
| rating INT| NOT NULL (1..5) |

```

```

| briefing_clear VARCHAR(1000)|
| educational_value VARCHAR(1000)|
| recommend VARCHAR(1000)|
| comment VARCHAR(1000)|
| routed_to VARCHAR(60)|
| submitted_at INSTANT| NOT NULL
| created_at INSTANT| NOT NULL
| updated_at INSTANT| NOT NULL
+-----+

+-----+
| survey_send
+-----+
| id BIGINT| PK
| visitor_id BIGINT| FK -> visitor (NOT NULL)|
| channel VARCHAR(20)| NOT NULL
| message_template VARCHAR(500)|
| sent_at INSTANT| NOT NULL
| status VARCHAR(20)| NOT NULL, 'SENT'|
| created_at INSTANT| NOT NULL
| updated_at INSTANT| NOT NULL
+-----+

+-----+
| tour_stop
+-----+
| id BIGINT| PK
| name VARCHAR(150)| NOT NULL
| position INT| NOT NULL
| description VARCHAR(2000)|
| duration_minutes INT| NOT NULL
| max_capacity INT|
| location VARCHAR(150)|
| demo VARCHAR(1000)|
| safety_notes VARCHAR(1000)|
| active BOOLEAN| NOT NULL, true
| created_at INSTANT| NOT NULL
| updated_at INSTANT| NOT NULL
+-----+

+-----+
| workshop
+-----+
| id BIGINT| PK
| name VARCHAR(150)| NOT NULL
| duration_minutes INT| NOT NULL
| target_group VARCHAR(60)| NOT NULL
| facilitator VARCHAR(100)|
| description VARCHAR(1000)|
| status VARCHAR(20)| NOT NULL, 'DRAFT'|
| created_at INSTANT| NOT NULL
| updated_at INSTANT| NOT NULL
+-----+

+-----+
| agri_activity
+-----+
| id BIGINT| PK
| name VARCHAR(150)| NOT NULL
| price DECIMAL(12,2)| NOT NULL
| capacity INT| NOT NULL
| duration_minutes INT| NOT NULL
| description VARCHAR(1000)|

```

```

| active      BOOLEAN| NOT NULL, true |
| created_at  INSTANT| NOT NULL      |
| updated_at  INSTANT| NOT NULL      |
+-----+

+-----+
|          booking          |
+-----+
| id            BIGINT| PK          |
| agri_activity_id BIGINT| FK NOT NULL|
| time_slot_id   BIGINT| FK NOT NULL|
| visitor_full_name VARCHAR(150)| NN|
| visitor_email  VARCHAR(255)| NN  |
| visitor_phone  VARCHAR(30)|      |
| people_count   INT| NOT NULL   |
| payment_method VARCHAR(20)|      |
| payment_status VARCHAR(20)| UNPAID |
| status         VARCHAR(20)| PENDING |
| reminder_scheduled_at INSTANT|      |
| created_at     INSTANT| NOT NULL |
| updated_at     INSTANT| NOT NULL |
+-----+

```

Table Summary

Table	Rows (design)	Purpose
`time_slot`	~500/year	Farm tour calendar slots (2/day max)
`visitor`	~2,000	Visitor database (individuals/groups)
`registration`	~4,000	Links visitor -> time slot **or** event
`briefing`	~3,000	Safety briefing per confirmed registration
`event`	~50/year	Open days, buyer visits, school/community days
`feedback`	~3,000	Satisfaction responses (tablet + survey links)
`survey_send`	~8,000	Post-visit surveys sent by email/SMS/WhatsApp
`tour_stop`	~8	Standard guided-tour stops (seeded via API)
`workshop`	~10	Tour templates / workshops (Active / Draft / Inactive)
`agri_activity`	~5	Paid agritourism offers (catalogue)
`booking`	~2,000/yr	Paid reservations linked to an activity + time slot

Enums

Enum	Values	Table Column
`TimeSlotStatus`	`AVAILABLE`, `RESERVED`, `FULL`, `CANCELLED`, `COMPLETED`	`time_slot.status`
`RegistrationStatus`	`PENDING`, `CONFIRMED`, `REJECTED`, `CHECKED_IN`, `CANCELLED`	`registration.status`
`BriefingStatus`	`PENDING`, `DONE`	`briefing.status`
`VisitorType`	`INDIVIDUAL`, `GROUP`, `SCHOOL`, `PARTNER`	`visitor.visitor_type`
`VisitPurpose`	`TOURISM`, `PURCHASE`, `PARTNERSHIP`, `INVESTMENT`, `EDUCATION`, `OTHER`	`registration.visit_purpose`
`EventType`	`OPEN_DAY`, `PARTNER_BUYER`, `SCHOOL`, `COMMUNITY`, `OTHER`	`event.event_type`
`EventStatus`	`DRAFT`, `PUBLISHED`, `CANCELLED`, `COMPLETED`	`event.status`
`FeedbackChannel`	`ON_SITE`, `EMAIL`, `SMS`, `WHATSAPP`	`feedback.origin`, `survey_send.channel`
`SurveyStatus`	`SENT`, `RECEIVED`	`survey_send.status`
`WorkshopStatus`	`DRAFT`, `ACTIVE`, `INACTIVE`	`workshop.status`
`BookingStatus`	`PENDING`, `CONFIRMED`, `COMPLETED`, `CANCELLED`	`booking.status`
`BookingPaymentStatus`	`UNPAID`, `PAID`, `REFUNDED`	`booking.payment_status`
`BookingPaymentMethod`	`CASH`, `ORANGE_MONEY`, `MOOV_MONEY`	`booking.payment_method`

3. REST API Endpoints

Base URL: <http://localhost:8080/api/v1>

3.1 Time Slots - /api/v1/time-slots

Method	Path	Description	Status Codes
GET	/time-slots	List all slots (optional `?date=YYYY-MM-DD`)	200
GET	/time-slots/{id}	Get slot by ID	200, 404
POST	/time-slots	Create a new slot	201, 400, 409, 422
PUT	/time-slots/{id}	Update slot details	200, 404, 422
DELETE	/time-slots/{id}	Cancel a slot (soft-delete)	204, 404, 422
POST	/time-slots/{id}/assign-guide	Assign a guide (`?guideId=N`)	200, 404, 422
GET	/time-slots/availability?date=...	Availability view for a date	200

```
{
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "guideId": 5
}
```

Create/Update Request Body:

```
{
  "id": 1,
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "booked": 3,
  "status": "RESERVED",
  "guideId": 5,
  "createdAt": "2026-09-01T10:00:00Z",
  "updatedAt": "2026-09-01T10:00:00Z"
}
```

TimeSlotResponse:

```
{
  "id": 1,
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "remaining": 7,
  "status": "RESERVED",
  "guideId": 5
}
```

AvailabilityResponse:

3.2 Visitors - /api/v1/visitors

Method	Path	Description	Status Codes
GET	/visitors	List all visitors	200
GET	/visitors/{id}	Get visitor by ID	200, 404
POST	/visitors	Create visitor	201, 400, 409
PUT	/visitors/{id}	Update visitor	200, 404, 409

```
{
  "fullName": "Alice Dupont",
  "groupSize": 2,
  "email": "alice@example.com",
  "phone": "+241 06 12 34 56",
  "language": "French",
  "type": "INDIVIDUAL",
  "specialNeeds": "Wheelchair access required"
}
```

```
}
```

VisitorRequest:

3.3 Registrations - /api/v1/registrations

Method	Path	Description	Status Codes
GET	/registrations	List (filter by `?timeSlotId=`, `?date=`, or `?prospect=true`)	200
GET	/registrations/{id}	Get registration by ID	200, 404
POST	/registrations	Create registration (visitor->slot, requires `visitPurpose`)	201, 400, 409, 422
PATCH	/registrations/{id}/approve	Approve -> triggers briefing creation	200, 404, 422
PATCH	/registrations/{id}/reject	Reject a pending registration	200, 404, 422
PATCH	/registrations/{id}/check-in	Check in a confirmed visitor	200, 404, 422
PATCH	/registrations/{id}/cancel	Cancel a registration	200, 404, 422

```
{
  "visitorId": 1,
  "timeSlotId": 10,
  "visitPurpose": "PURCHASE",
  "eventId": null
}
```

RegistrationRequest:

```
{
  "id": 1,
  "visitorId": 1,
  "visitorName": "Alice Dupont",
  "groupSize": 2,
  "timeSlotId": 10,
  "slotDate": "2026-09-08",
  "slotStart": "09:00",
  "eventId": null,
  "visitPurpose": "PURCHASE",
  "prospect": true,
  "status": "PENDING",
  "briefingId": null,
  "createdAt": "2026-09-01T10:00:00Z",
  "updatedAt": "2026-09-01T10:00:00Z"
}
```

RegistrationResponse (new fields):

3.4 Briefings - /api/v1

Method	Path	Description	Status Codes
GET	/registrations/{registrationId}/briefing	Get briefing for registration	200, 404
PATCH	/registrations/{registrationId}/briefing/deliver	Mark briefing delivered	200, 422

```
{
  "staffMember": "Guard Camille",
  "signature": "digital-signature-or-name"
}
```

BriefingDeliverRequest:

3.5 Events - /api/v1/events

Method	Path	Description	Status Codes
GET	/events	List events (`?type=`, `?date=YYYY-MM-DD`)	200
GET	/events/{id}	Get event by ID	200, 404
POST	/events	Create event (default status `DRAFT`)	201, 400, 422
PUT	/events/{id}	Update event (not allowed when CANCELLED/COMPLETED)	200, 404, 422
DELETE	/events/{id}	Cancel event (soft-delete)	204, 404, 422
POST	/events/{id}/publish	Publish a DRAFT event	200, 404, 422

Method	Path	Description	Status Codes
GET	/events/{id}/registrations	List registrations for the event	200, 404
POST	/events/{id}/register	Register a visitor for the event	201, 400, 404, 409, 422

```
{
  "title": "Open Farm Day",
  "type": "OPEN_DAY",
  "startDateTime": "2026-10-04T10:00:00",
  "endDateTime": "2026-10-04T16:00:00",
  "maxCapacity": 60,
  "description": "Discover the farm, meet the team, taste our products",
  "location": "Farm entrance hall"
}
```

EventRequest:

```
{
  "id": 1,
  "title": "Open Farm Day",
  "type": "OPEN_DAY",
  "startDateTime": "2026-10-04T10:00:00",
  "endDateTime": "2026-10-04T16:00:00",
  "maxCapacity": 60,
  "booked": 3,
  "status": "PUBLISHED",
  "description": "Discover the farm, meet the team, taste our products",
  "location": "Farm entrance hall",
  "createdAt": "2026-09-01T10:00:00Z",
  "updatedAt": "2026-09-01T10:00:00Z"
}
```

EventResponse:

```
{
  "visitorId": 1,
  "visitPurpose": "PURCHASE"
}
```

EventRegistrationRequest:

The registration created for an event has timeSlotId: null and carries the visitor's visitPurpose. Commercial purposes (PURCHASE, PARTNERSHIP, INVESTMENT) automatically mark the visitor as a prospect, exactly like time-slot registrations.

3.6 Feedback & Surveys - /api/v1

Method	Path	Description	Status Codes
GET	/feedback	List responses (?visitorId=, ?from=, ?to=)	200
GET	/feedback/summary	Compiled stats (?from=, ?to=)	200
POST	/feedback	Submit response (on-site or linked to a sent survey)	201, 400, 404, 422
PATCH	/feedback/{id}/route	Route a response to a module owner	200, 404, 400
GET	/surveys	List sent surveys (?visitorId=, ?status=)	200
POST	/surveys	Send a post-visit survey (email/SMS/WhatsApp)	201, 400, 404, 422

```
{
  "visitorId": 1,
  "surveyId": 12,
  "rating": 4,
  "briefingClear": "Yes, very clear",
  "educationalValue": "The solar tracking was interesting",
  "recommend": "Definitely",
  "comment": "Would love a German-language brochure"
}
```

FeedbackRequest: visitorId (required), surveyId (optional - links the response to a sent survey and marks it RECEIVED), rating (1-5, required), briefingClear, educationalValue, recommend, comment.

```
{
  "id": 1,
  "visitorId": 1,
  "visitorName": "Alice Dupont",
  "surveyId": 12,
  "origin": "EMAIL",
  "rating": 4,
  "briefingClear": "Yes, very clear",
  "educationalValue": "The solar tracking was interesting",
  "recommend": "Definitely",
  "comment": "Would love a German-language brochure",
  "routedTo": "Sales (Mariata)",
  "submittedAt": "2026-08-26T13:00:00Z",
  "createdAt": "2026-08-26T13:00:00Z",
  "updatedAt": "2026-08-26T13:00:00Z"
}
```

FeedbackResponse:

```
{
  "averageRating": 4.2,
  "total": 18,
  "recommendPct": 92.0,
  "distribution": { "1": 0, "2": 0, "3": 3, "4": 6, "5": 9 }
}
```

FeedbackSummaryResponse (compiled weekly report):

```
{
  "visitorId": 3,
  "channel": "EMAIL",
  "messageTemplate": "Thank you for visiting Sustainable Farm! Share your feedback: [link]"
}
```

SurveyRequest:

SurveyResponse: id, visitorId, visitorName, channel, messageTemplate, sentAt, status (SENT/RECEIVED), rating (once the visitor responded), createdAt, updatedAt.

On-site tablet submissions use POST /feedback without surveyId -> origin: "ON_SITE". A sent survey becomes RECEIVED automatically when a response is linked to it.

3.7 Educational Program - /api/v1

Method	Path	Description	Status Codes
GET	/tour-stops	List guided-tour stops (ordered by position)	200
POST	/tour-stops	Create a tour stop	201, 400, 409
PUT	/tour-stops/{id}	Update a stop (position must stay unique among active stops)	200, 404, 409, 400
DELETE	/tour-stops/{id}	Deactivate a stop (soft-delete)	204, 404
GET	/workshops	List workshops/templates (?status=DRAFT ACTIVE INACTIVE)	200
POST	/workshops	Create a workshop (status DRAFT)	201, 400
PUT	/workshops/{id}	Update a workshop	200, 404, 400
POST	/workshops/{id}/publish	Publish a DRAFT workshop -> ACTIVE	200, 404, 422
POST	/workshops/{id}/deactivate	Deactivate an ACTIVE workshop -> INACTIVE	200, 404, 422

```
{
  "name": "Solar plant & tracking system",
  "position": 4,
  "description": "86.4 kWp installed, +20% yield with tracking",
  "durationMinutes": 15,
  "maxCapacity": 15,
  "location": "Solar field - observation path",
  "demo": "Live dashboard screen (Infineon sensors)",
  "safetyNotes": "Observation path only; tracker, battery/inverter and running generator restricted"
}
```


TourStopRequest / TourStopResponse: name (required), position (required, ?1, unique among active stops), description, durationMinutes (required, ?1), maxCapacity (optional, ?1), location, demo, safetyNotes, active.

WorkshopRequest / WorkshopResponse: name (required), durationMinutes (required, ?1), targetGroup (required, e.g. All, Professional, Schools), facilitator, description, status (DRAFT by default).

Confirmed reference content (module owners: Aida, Jean-Louis, Delwende, Phares, Abdoul): standard tour stops - Welcome & safety briefing (5-10 min, Front Desk), Mango orchard (20 min, Keitt 200 trees/2 ha), Smart drip irrigation (10-15 min, max 10), Solar plant & tracking (15 min, max 15), Processing unit (20 min, max 10), Wrap-up & questions (15 min) -> max 10 pers/slot (limiting stops: irrigation + processing). Reference workshops: Standard farm tour (~100 min, All, Alix), Solar workshop (45 min, Professional, Guest Aida), Mango tasting & processing (40 min, General public, Guest Abdoul), School discovery day (120 min, Schools, P. Nikiéma).

3.8 Agritourism Booking System - /api/v1

Method	Path	Description	Status Codes
GET	/activities	List agritourism activities (alphabetical)	200
POST	/activities	Create an activity (name, price FCFA, capacity, durationMinutes)	201, 400
PUT	/activities/{id}	Update an activity	200, 404, 400
DELETE	/activities/{id}	Deactivate an activity (soft-delete)	204, 404
GET	/bookings	List bookings (?status=PENDING CONFIRMED COMPLETED CANCELLED, ?activityId=, ?date=)	200
GET	/bookings/occupancy	Occupancy % per activity	200
POST	/bookings	Create a booking (validates capacity)	201, 400, 404, 409
PUT	/bookings/{id}	Update a booking (PENDING/CONFIRMED only)	200, 404, 409, 422
POST	/bookings/{id}/pay	Record payment -> PAID	200, 404, 422
POST	/bookings/{id}/confirm	Confirm (only PENDING + PAID), schedules 24h reminder	200, 404, 422
POST	/bookings/{id}/complete	Mark a CONFIRMED booking COMPLETED	200, 404, 422
POST	/bookings/{id}/cancel	Cancel PENDING/CONFIRMED; paid -> REFUNDED; frees capacity	200, 404, 422

```
{
  "name": "Solar workshop",
  "price": 5000,
  "capacity": 15,
  "durationMinutes": 45,
  "description": "Guided visit of the 86.4 kwp solar plant with tracking"
}
```

AgriActivityRequest / AgriActivityResponse: name (150), price (BigDecimal ? 0, FCFA), capacity (? 1), durationMinutes (? 1), description (1000), active.

```
{
  "activityId": 1,
  "timeSlotId": 2,
  "visitorFullName": "Lucas Weber",
  "visitorEmail": "lucas@example.com",
  "visitorPhone": "+22612345678",
  "peopleCount": 5,
  "paymentMethod": "ORANGE_MONEY"
}
```

BookingRequest / BookingResponse: activityId, timeSlotId, visitorFullName, visitorEmail (valid email), visitorPhone, peopleCount (? 1), paymentMethod (CASH, ORANGE_MONEY, MOOV_MONEY). Response adds reference (BK-00001 format), activityName, activityPrice, slot date/times, totalAmount (price × people), paymentStatus, status, reminderScheduledAt.

Reference data from the mockup: Standard tour 5,000 FCFA/person (family rate 3,500 FCFA); payment accepted in cash, Orange Money, Moov Money. Booking flow: choose slot & tour type -> registration form -> email confirmation (mockup only) -> reminder 24 h before (computed on confirmBooking, actual email delivery is a documented follow-up).

4. Business Rules

Scheduling Rules (SchedulingRules.java)

Rule	Value	Source
Max visitors per slot	**10**	Confirmed with module owners (Abdoul)
Max slots per day	**2**	Mockup: morning + afternoon
Closed day	**Sunday**	All module owners
Morning slot	09:00-11:00	Default
Afternoon slot	14:00-16:00	Default

Registration Rules

Rule	Enforcement
Cannot register on a **CANCELLED** slot	`BusinessRuleException`
No duplicate registration (visitor + slot)	`ConflictException` (409)
Group size must fit remaining capacity	`BusinessRuleException`
Only **PENDING** -> approve/reject	`BusinessRuleException`
Only **CONFIRMED** -> check-in	`BusinessRuleException`
Cannot cancel after **CHECKED_IN**	`BusinessRuleException`
CONFIRMED -> auto-creates `Briefing` (PENDING)	Service layer
Briefing delivery requires PENDING status	`BusinessRuleException`

Event Rules

Rule	Enforcement
`endTime` must be after `startTime`	`BusinessRuleException`
Only **DRAFT** events can be published	`BusinessRuleException`
Cannot publish an event that has already ended	`BusinessRuleException`
Cannot update **CANCELLED** or **COMPLETED** events	`BusinessRuleException`
Cannot cancel a **COMPLETED** event	`BusinessRuleException`
Registration only on **PUBLISHED** events	`BusinessRuleException`
No duplicate registration (visitor + event)	`ConflictException` (409)
`booked + groupSize > maxCapacity` -> reject	`BusinessRuleException` (422)
Booked count excludes `REJECTED` / `CANCELLED` registrations	Repository query

Feedback & Survey Rules

Rule	Enforcement
Rating must be between 1 and 5	Bean validation (400)
Response linked to a survey must match the survey's visitor	`BusinessRuleException` (422)
A survey can only be answered once (RECEIVED is final)	`BusinessRuleException` (422)
Survey channel cannot be `ON_SITE` (tablet responses are direct)	`BusinessRuleException` (422)
Response without a survey -> recorded as `ON_SITE`	Service layer
Linking a response marks the survey `RECEIVED`	Service layer
Routing targets are free-text tags (no coupling to other modules)	Data model

Educational Program Rules

Rule	Enforcement
Position must be ? 1	Bean validation (400)
Position must be unique among **active** stops	`ConflictException` (409)
Deactivating a stop frees its position (soft-delete)	`DELETE /tour-stops/{id}`
Create -> status `DRAFT`	Service layer
Only **DRAFT** workshops can be published	`BusinessRuleException` (422)
Only **ACTIVE** workshops can be deactivated	`BusinessRuleException` (422)

Agritourism Booking Rules

Rule	Enforcement
Capacity counts **non-cancelled** bookings only	Repository aggregate query

```
| A booking can never exceed the **slot capacity** (10 pers) | `ConflictException` (409) |
| A booking can never exceed the **activity capacity** | `ConflictException` (409) |
| Bookings only on **active** activities and non-cancelled slots | `BusinessRuleException` (422) |
| Cannot update **COMPLETED / CANCELLED** bookings | `BusinessRuleException` (422) |
| Can only confirm a **PENDING** booking that is **PAID** | `BusinessRuleException` (422) |
| Confirm schedules the 24 h reminder (`slot start ? 24h`) | Service layer |
| Completing requires **CONFIRMED** status | `BusinessRuleException` (422) |
| Cancelling a **PAID** booking marks it **REFUNDED** | Service layer |
```

Visitor Rules

```
| Rule | Enforcement |
|-----|-----|
| Duplicate email -> `ConflictException` (409) | On create and update |
| `@NotBlank` on fullName | Bean validation (400) |
| `@Email` on email | Bean validation (400) |
```

Lead Generation (Prospect Detection)

The visitor management app doubles as a marketing tool for the Sales & Marketing team. When a visitor registers with a commercial purpose, the system automatically flags them as a prospect.

How it works:

1. During registration, the visitor must select a visitPurpose (mandatory field)
2. If the purpose is PURCHASE, PARTNERSHIP, or INVESTMENT, the system sets isProspect = true automatically
3. Sales & Marketing can query all prospects via GET /api/v1/registrations?prospect=true

VisitPurpose values:

```
| Purpose | Prospect? | Description |
|-----|-----|-----|
| `TOURISM` | No | Leisure visit, tourism |
| `PURCHASE` | **Yes** | Interested in buying products |
| `PARTNERSHIP` | **Yes** | Business partnership opportunity |
| `INVESTMENT` | **Yes** | Investment opportunity |
| `EDUCATION` | No | Training, educational visit |
| `OTHER` | No | Other reasons |
```

```
GET /api/v1/registrations?prospect=true
```

Prospect query example: Returns all registrations where isProspect = true, including visitor name, email, phone, and visit purpose.

5. Status Flow Diagrams

TimeSlot Status Flow

```

      +-----+
      | create | AVAILABLE |
      +-----+----->
                        |
                        +-----?-----+
                        |               |
                    register    assign    cancel
                        |       guide    |
                        v               v
      +-----+       |       +-----+
      | RESERVED |       |       | CANCELLED |
      +-----+       |       +-----+
                        |               |
                    full |               |
                        v               |
      +-----+       |               |
      | FULL |<-----+ (booked >= capacity)
      +-----+
      |

```

```

end_time passes
  v
+-----+
| COMPLETED |
+-----+

```

Registration Status Flow

```

      +-----+
register | PENDING |
-----> +-----+
      |
      +-----?-----+
      |       |       |
  approve reject  cancel
      |       |       |
      v       v       v
+-----+ +-----+ +-----+
| CONFIRMED | | REJECTED | | CANCELLED |
+-----+ +-----+ +-----+
      |
  check-in
      |
      v
+-----+
| CHECKED_IN |
+-----+
      |
  cancel blocked
      |
  (final state)

```

Briefing Status Flow

```

      +-----+
auto   | PENDING | <- created when registration CONFIRMED
-----> +-----+
      |
  deliver briefing
      |
      v
+-----+
|  DONE  | <- staffMember + signature + deliveredAt
+-----+

```

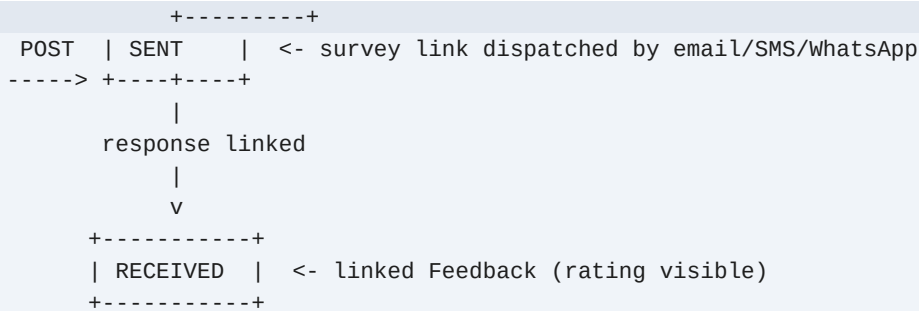
Event Status Flow

```

      +-----+
create  | DRAFT  |
-----> +-----+
      |
  publish (not ended)
      |
      v
+-----+          cancel from any non-completed
| PUBLISHED |<-----+
+-----+          |
      |          +-----+
  ended |          | CANCELLED |
      |          +-----+
      v
+-----+
| COMPLETED |
+-----+

```

Survey Status Flow



6. Error Handling

All errors return a consistent JSON envelope via `GlobalExceptionHandler`:

```
{
  "status": 404,
  "error": "Not Found",
  "message": "Time slot 999 not found",
  "path": "/api/v1/time-slots/999"
}
```

Exception	HTTP Status	When
<code>`ResourceNotFoundException`</code>	404	Entity not found
<code>`ConflictException`</code>	409	Duplicate email, duplicate registration, overlapping slot
<code>`BusinessRuleException`</code>	422	Invalid state transition, capacity exceeded, Sunday creation
<code>`MethodArgumentNotValidException`</code>	400	Bean validation failures (<code>@NotBlank`</code> , <code>@Email`</code> , etc.)

7. Test Coverage Summary

Layer	Annotation	Tests	Framework
Repository	<code>`@DataJpaTest`</code> + H2	38	Spring Data + AssertJ
Service	<code>`@ExtendWith(MockitoExtension.class)`</code>	91	Mockito + AssertJ
Controller	<code>`@WebMvcTest`</code> + MockMvc	82	MockMvc + Mockito
Context	<code>`@SpringBootTest`</code>	1	Spring Boot
Total		**212**	

Test Files

```
src/test/java/com/infineonbit/sustainablefarm/
+-- SustainableFarmApplicationTests.java
+-- modules/visitormanagement/
    +-- repository/
        | +-- TimeSlotRepositoryTest.java      (5 tests)
        | +-- VisitorRepositoryTest.java       (4 tests)
        | +-- RegistrationRepositoryTest.java  (5 tests)
        | +-- BriefingRepositoryTest.java      (2 tests)
        | +-- EventRepositoryTest.java         (4 tests)
        | +-- FeedbackRepositoryTest.java      (7 tests)
        | +-- EducationalProgramRepositoryTest.java (5 tests)
        | +-- AgriTourismRepositoryTest.java   (6 tests)
    +-- service/
        | +-- SchedulingServiceImplTest.java   (13 tests)
        | +-- RegistrationServiceImplTest.java (21 tests)
        | +-- EventServiceImplTest.java        (16 tests)
        | +-- FeedbackServiceImplTest.java     (12 tests)
        | +-- EducationalProgramServiceImplTest.java (12 tests)
        | +-- BookingServiceImplTest.java      (17 tests)
    +-- controller/
        | +-- TimeSlotControllerTest.java      (9 tests)
        | +-- VisitorControllerTest.java       (7 tests)
```

```

+-- RegistrationControllerTest.java    (12 tests)
+-- BriefingControllerTest.java        (5 tests)
+-- EventControllerTest.java           (13 tests)
+-- FeedbackControllerTest.java        (10 tests)
+-- EducationalProgramControllerTest.java (12 tests)
+-- BookingControllerTest.java         (13 tests)

```

8. Running the Backend

```

# Start PostgreSQL + Backend
docker compose up -d backend

# Backend available at
http://localhost:8080

# Swagger UI
http://localhost:8080/swagger-ui/index.html

# Actuator health
http://localhost:8080/actuator/health

# Run tests (Docker, no local JDK required)
docker run --rm -v "$PWD":/src -w /src \
  -v "$HOME/.m2":/root/.m2 \
  maven:3.9-eclipse-temurin-21 ./mvnw test

# Compile only
docker run --rm -v "$PWD":/src -w /src \
  -v "$HOME/.m2":/root/.m2 \
  maven:3.9-eclipse-temurin-21 ./mvnw compile

```

9. Technology Stack

Component	Version	Notes
Java	21	LTS
Spring Boot	4.1.0	Modular monolith
Spring Data JPA	4.1.0	Derived queries + `@Query`
Hibernate ORM	7.4.1	`ddl-auto=update` (no Flyway)
PostgreSQL	17-alpine	Production database
H2	2.x	Test-only in-memory DB
SpringDoc OpenAPI	2.8.5	Swagger UI auto-generated
Lombok	1.18.46	Boilerplate reduction
Jackson	3.1.4	JSON serialization
Mockito	5.23.0	Service + controller tests
AssertJ	3.27.7	Fluent assertions

Last updated: 2026-09-09